

BSc

ROBOTICS

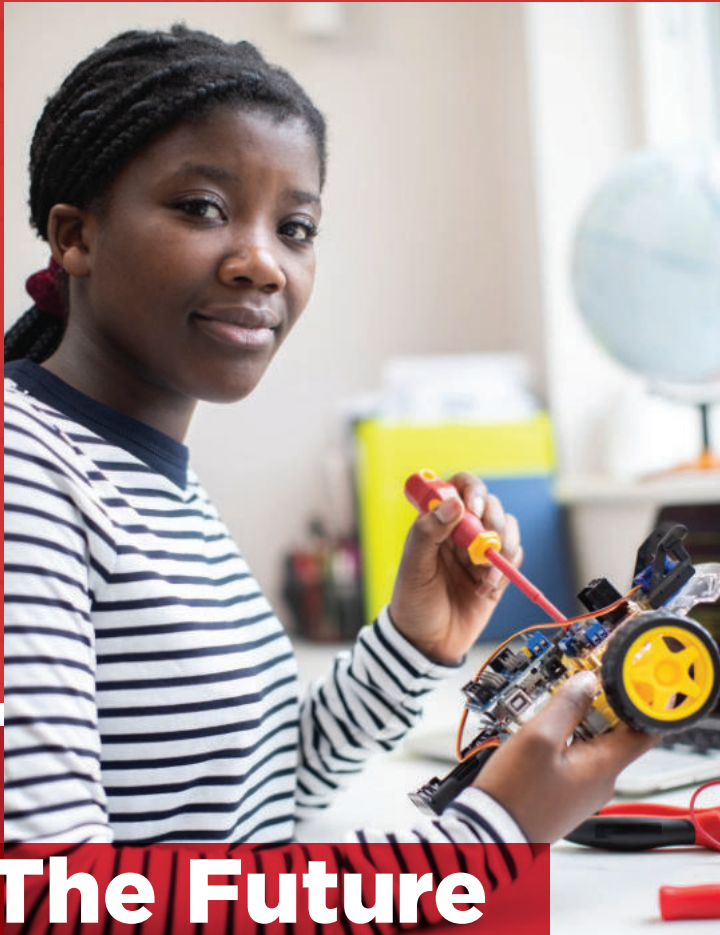
ENGINEERING

Robotics Engineering at ACity

ACity's groundbreaking, BSc Robotics Engineering program leads the efforts at developing a cadence of engineering programs that will push the boundaries of knowledge while establishing a foothold for students in jobs-of-the-future.

In this program, emphasis is placed on design, analysis and modeling, programming, integration of Artificial Intelligence (AI), manufacturing, usage and management of robots. Additionally, the program is deliberately designed to play comfortably at the intersection of AI and Biomedical providing further impetus to new programs in Data Analytics and Advanced Autonomous Systems.

Our goal is to create a versatile graduate conversant in a broad area of Robotics and poised to adapt swiftly to the rapidly evolving field of robotics.



The Future is Robotics!

Gone are the days when the role of assistive devices in human life was pure fiction and tales. Today, the fast-evolving field of robotics has become an integral part of human life.

As the African continent strives for industrialization, it is important to note that efficient, productive yet cheaper work strategies across key areas such as manufacturing, agriculture, education, healthcare, among others, shall be pivoted around automated devices.

According to a publication on www.techjury.net on July 28, 2020, 88% of businesses worldwide plan to adopt robotic automation into their infrastructure, with robotics already employing about 150,000 people worldwide in engineering and assembly jobs.

Between 2017 and 2020, the number of industrial robots alone installed across industries worldwide could exceed a total of 3.05 million, revealed by the Global Robotics Industry statistics.

Africa's advancement in robotics means grooming local experts, who understand the peculiar needs of the continent, and can apply the knowledge perfectly to build robots that can perform high-risk and complex tasks.

Top Careers in Robotics

Wondering what unique opportunities are available for you in the world of work after a 4-year of exciting, explorative and hands-on Robotics Engineering undergraduate program at ACity?

Here are some top careers in the field:

- Automated Product Design Engineer
- R&D Product Development Engineer
- Quality Control Engineer
- Front End Developer
- Hardware Engineer
- Robotics Engineer
- Software Engineer
- CAD Engineer

Why ACity?

Home to
students from over
23 countries
across Africa

A fully digitized
campus with the fastest
internet speed
in a Ghanaian university

70%
practical lessons

**Experienced
Faculty**
with Industry Expertise

Every **ACitizen** has
the opportunity to do a
work placement
every vacation

Student-To-Faculty Ratio
5:1

Houses a
Digital Makerspace
and a **Technology &
Entrepreneurial Center**
for **Creative freedom
& Innovation**

Our Unique Learning Pillars



Experiential Learning

Hands-on learning to prepare students to readily apply concepts, to easily integrate into the workspace.



Contextual Learning

Solving real grass-root problems to expose students to the local context and develop empathy towards the continent's progress.



Unified Learning

A project-based approach that combines concepts across courses to connect the dots and enable unified learning.



Extensional Learning

Arms students with a viable toolkit to help them confront real-life issues, they may not have encountered during their academic life, squarely.

Entry Requirements

WASSCE/NECO

Minimum C6 in 6 subjects including 3 core subjects (English, Mathematics, Integrated/General Science) and 3 elective subjects. (Physics, Elective Mathematics + Chemistry OR any other elective relevant to the chosen Programme)

SSCE

Minimum D or a pass in 6 subjects including 3 core subjects (English, Mathematics, Integrated/General Science) and 3 elective subjects. (Physics, Elective Mathematics + Chemistry OR any other elective relevant to the chosen Programme)

IGCSE O-LEVEL & A-LEVEL

Minimum of 5 credit passes in the IGCSE/O-Levels (English and Maths mandatory) and 3 passes in the A-Levels. (Elective/Add/Further Maths and Physics mandatory)

ENGLISH IB

Minimum of 5 credit passes in the IGCSE/O-Levels (English and Maths mandatory) and a minimum score of 4 points in 3 Higher Level (HL) subjects. (Elective/Add/Further Maths and Physics mandatory)

FRENCH IB

Minimum of 50% overall average pass. (subject to approval NAB) (Maths, English and Physics mandatory)

AMERICAN HIGH SCHOOL

Minimum GPA of 3.0 (Maths, English and Physics mandatory)

COURSE OUTLINE

SEMESTER 1

Course Name

Communication Skills

French Language

Fundamentals of Innovation and Entrepreneurship (FIE) Seminar I

Introduction to Engineering

Introduction to Programming with Python

Physical Sciences

Technology and Society

Pre-Calculus (with MATLAB)

SEMESTER 2

Course Name

African Studies

Fundamentals of Innovation and Entrepreneurship (FIE) Seminar II

Logic and Critical Thinking

Text and Meaning

Introduction to Multidisciplinary Design

Programming in C

Analytic Geometry and Calculus I (with MATLAB)

Circuit Theory

Engineering Mechanics

SEMESTER 3

Course Name

Fundamentals of Innovation and Entrepreneurship (FIE) I

Leadership Seminar I

Introduction to Material Science and Engineering

Analytic Geometry and Calculus II (with MATLAB)

Artificial Intelligence and Robotics

Basic Electronics

Computer-Aided Design and Manufacturing (CAD/CAM)

Mechanics of Materials

SEMESTER 4

Course Name

Fundamentals of Innovation and Entrepreneurship (FIE) II

Sensors, Measurements and Instrumentation

Applied Linear Algebra (with MATLAB)

Probability, Statistics and Reliability (with MATLAB)

Analogue and Digital Electronics

Fluid Mechanics

Unified Robotics I: Actuation

SEMESTER 5

Course Name

Leadership Seminar II
System Dynamics
Differential Equations and Numerical Methods (with MATLAB)
Machine Design
Microprocessors and Microcontrollers
Signals and Systems
Unified Robotics II: Sensing
Wireless Sensor Networks

SEMESTER 6

Course Name

Project Management, Engineering Economics and Risk Analysis
Automatic Control Systems
Digital Signal Processing
Embedded Systems
Machine Learning
Machines and Drives
Unified Robotics III: Manipulation

SEMESTER 7

Course Name

Advanced Instrumentation
Autonomous Agents
Computer Vision
Industrial Robotics
Industry Internship
Project Phase I
Unified Robotics IV: Navigation

SEMESTER 8

Course Name

Environmental Science and Engineering
Professional Ethics and Values
Modelling and Analysis of Mechatronic Systems
Project Phase II
Real-Time Digital Signal Processing